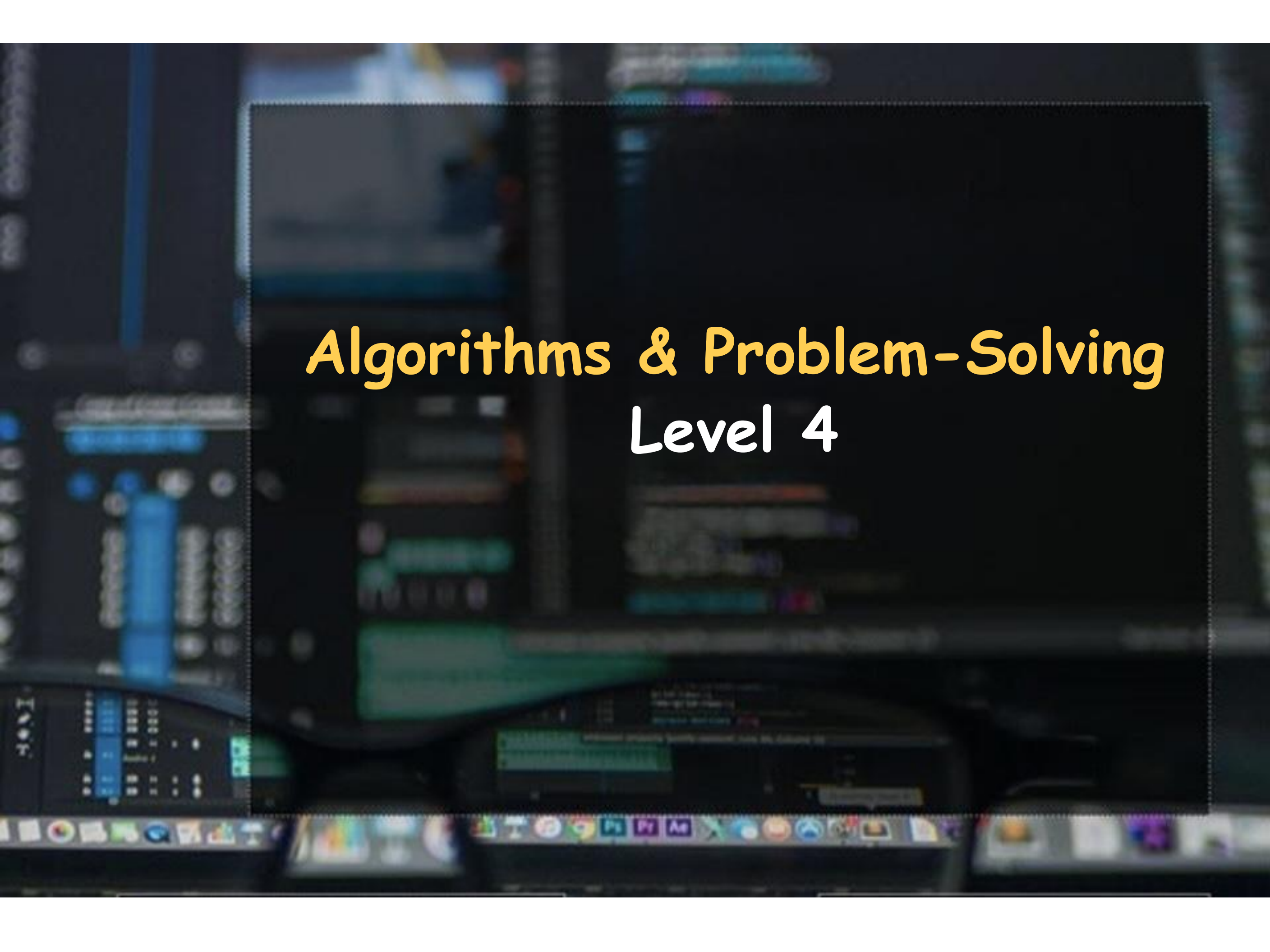
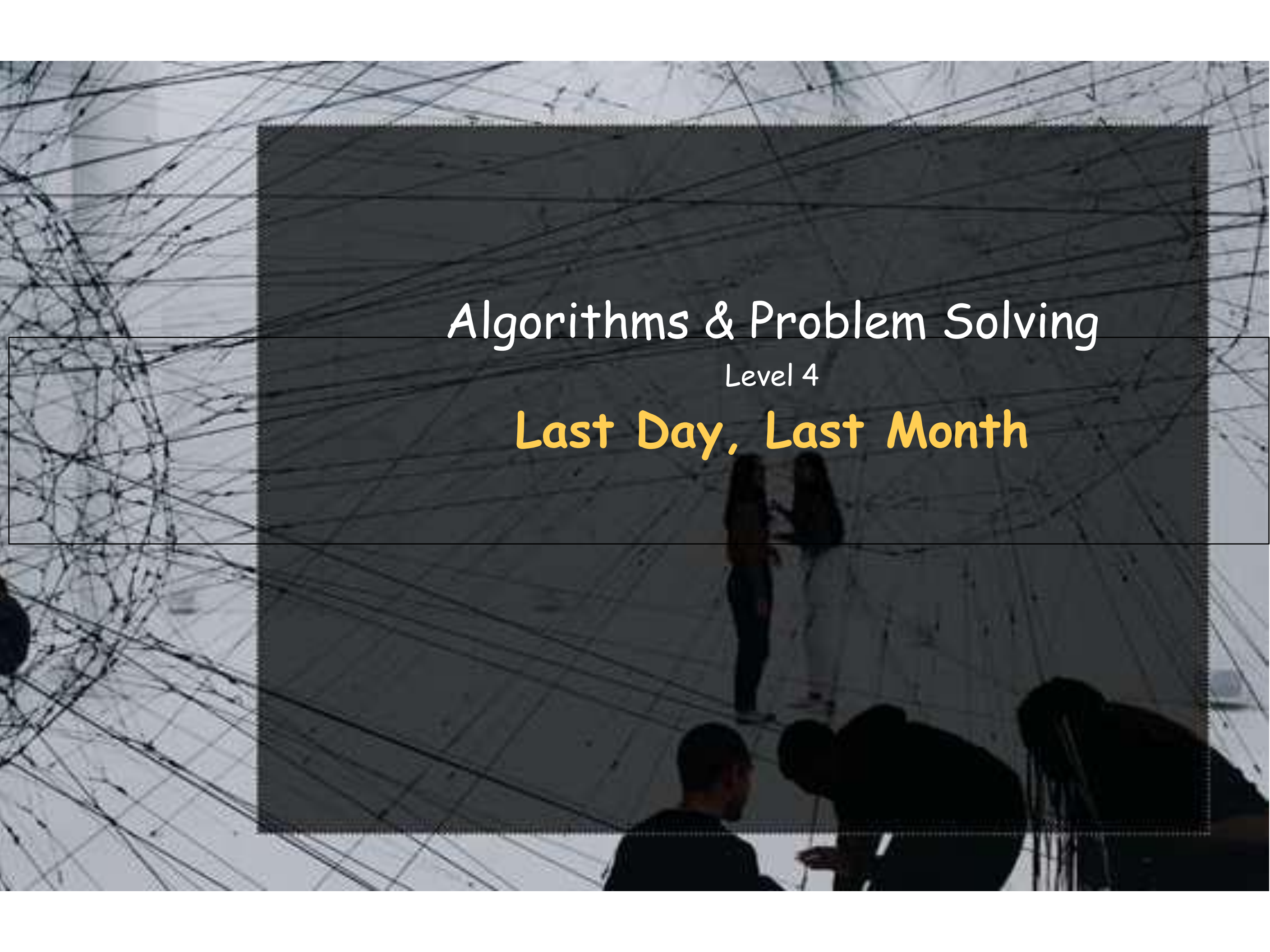




Algorithms & Problem-Solving Level 4



Algorithms & Problem Solving

Level 4

Last Day, Last Month



Definition

- The goal is a program that reads a date (day, month, year) and then checks:
 - Is today the last day of the month?
 - Is this month the last month of the year?



Explanation

- The program uses a struct to store the date (day, month, year).
- It then reads the values from the user using separate functions for the day, month, and year.
- Next, it checks:
 - If the year is a leap year.
 - The number of days in the month.
- Finally, it compares the day and month to determine if they are at the end of the month or the year.

Procedures

- 1 Define the `sDate` structure containing Year, Month, and Day
- 2 Create functions to input the day, month, and year from the user
- 3 Create an ``IsLeapYear()`` function to check if the year is a leap year
- 4 Create an ``NumberOfDaysInAMonth()`` function to determine the number of days in each month
- 5 Create an ``IsLastDayInMonth()`` function to check if today is the last day of the month
- 6 Create an ``IsLastMonthInYear()`` function to check if the month is December
- 7 In ``main()``:
 -  Read the date from the user
 -  Check if it is the last day of the month
 -  Check if it is the last month of the year
 -  Print the appropriate results



Example:

Please enter a Day? 31

Please enter a Month? 12

Please enter a Year? 2022

Yes, Day is Last Day in Month.

Yes, Month is Last Month in Year.



Explanation:

- The program is based on clear logic:
 - The last day of the month is when the total number of days in that month equals 12.
 - The last month of the year is when the total number of days equals 12.



Note:

- February has 29 days in a leap year and 28 in a regular year.
- The ``IsLeapYear()`` function automatically determines this.
- The ``NumberOfDaysInAMonth()`` function handles leap years correctly, so February will have 29 days instead of 28.
- These functions can be reused in other date-related programs, such as calculating the next day or the number of days between two dates.

Warning:

-  If an incorrect month is entered (less than 1 or greater than 12), the `NumberOfDaysInAMonth` function returns the value 0

Summary

- ✓ The program checks the last day of the month and the last month of the year.
- ✓ It relies on simple logic and helper functions.
- ✓ The program combines time logic with conditional checking.
- ✓ Using discrete functions makes the code organized and easy to read.
- ✓ Useful for understanding complex condition checking in C++ programming.

Thank you 😊